Weekly Report - Week 2

Semester 2, 2025
Date: August 10, 2025
Project: OAS - Oral Assessment System
Epic: 4.1 Academic Structure Management

1. Work Completed

This week I implemented Epic 4.1 (Academic Structure Management) for OAS. The goal was to support academic years, semesters and subjects with clean CRUD, validation, and access control.

Code branch: Epic4.1-Dev

Database

- Added three tables: `academic_year`, `semester`, `subject`.
- No database foreign keys; relationships are validated in the service layer.
- Seeded realistic sample data (years, semesters, subjects).

Backend

- Entities: `AcademicYear`, `Semester`, `Subject` with MyBatis-Plus mappings.
- Mappers: BaseMapper interfaces and XML stubs.
- Services: validation, uniqueness checks, and simple transaction use; logical relationship checks between entities.
- Controllers: REST CRUD with `@PreAuthorize`, `@Log`, pagination and filters.
- DTOs: request/response validation.

Exception Handling

- Extended `BusinessError` with 4001–4024 for the above modules.
- Use `BusinessException` + `GlobalExceptionHandler`; removed generic `RuntimeException`.

Key decisions

- No DB-level foreign keys; avoid coupling and cascades.
- Enforce consistency in services with logical ID references.

API Endpoints Summary

```
Academic Years:
GET    /api/academic-years          # Paginated list
GET    /api/academic-years/active   # Active years only
GET    /api/academic-years/{id}     # Get by ID
POST   /api/academic-years          # Create new
PUT    /api/academic-years/{id}     # Update
DELETE /api/academic-years/{id}     # Soft delete

Semesters:
GET    /api/semesters               # Paginated list
GET    /api/semesters/active        # Active semesters
GET    /api/semesters/by-academic-year/{id} # By academic year
POST   /api/semesters               # Create new
PUT    /api/semesters/{id}          # Update
DELETE /api/semesters/{id}          # Soft delete

Subjects:
GET    /api/subjects                # Paginated list
GET    /api/subjects/active         # Active subjects
GET    /api/subjects/my-subjects    # Current admin's subjects
GET    /api/subjects/by-semester/{id} # By semester
POST   /api/subjects                # Create new
PUT    /api/subjects/{id}           # Update
DELETE /api/subjects/{id}           # Soft delete
```

2. Next Week Plan

Epic 4.2: Assignment Management

- Design the assignment entity (supports individual/group/mixed types).
- Link assignments to subjects with validations.

- CRUD APIs consistent with 4.1.

Epic 4.3: Rubric Foundation

- Design a basic rubric structure and plan links to assignments.
- Explore options for hierarchical criteria where needed.

Continuous Improvement

- Keep code quality and patterns consistent.
- Add tests and docs alongside changes.

Total Effort: ~8 hours
Status: Epic 4.1 complete; moving to 4.2